# Chypnosis: Undervolting-based Static Side-Channel Attacks

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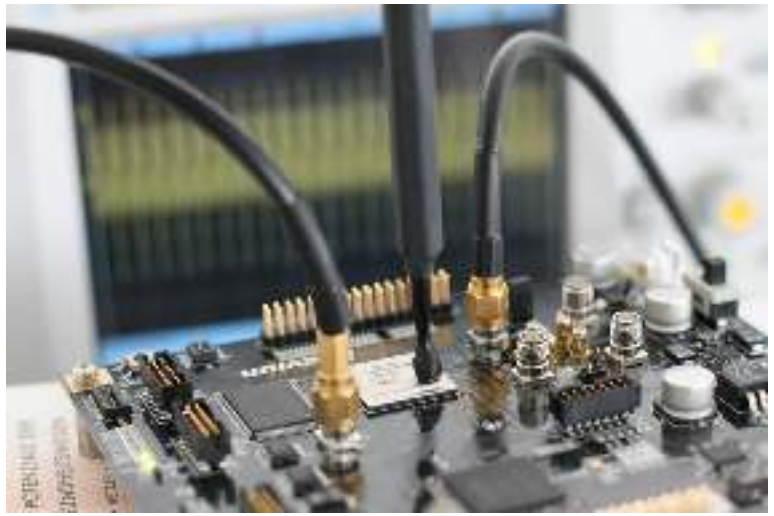
To appear at IEEE S&P 2026

Real World Crypto 2026

# Physical Side-channel Attacks

Fully-invasive Attacks



**Main assumption: Leakage happens during data/state transitions!**

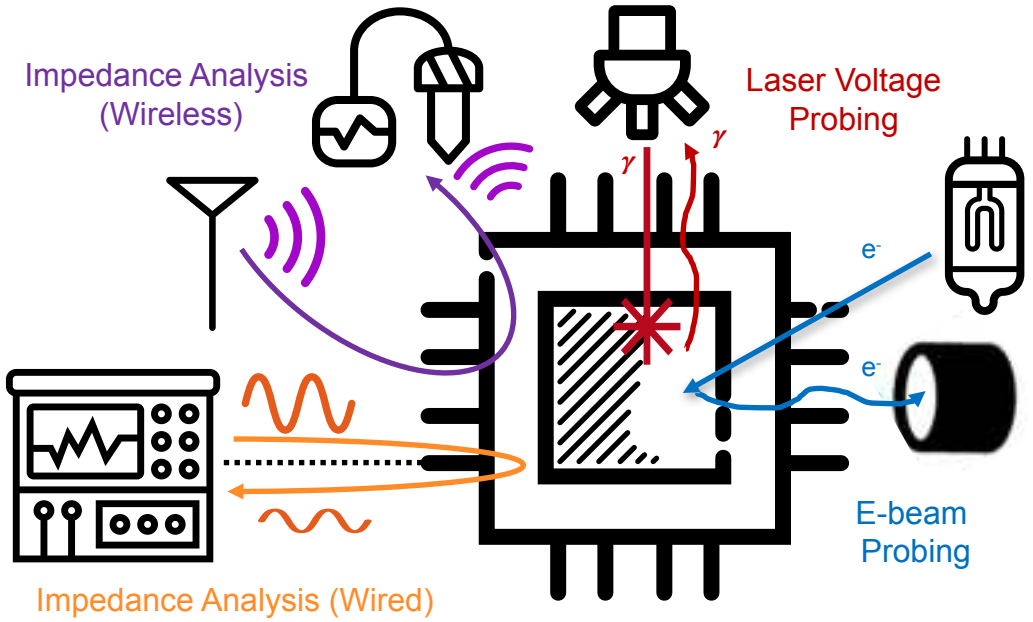


Image sources: Fraunhofer AISEC/TUM, Kleindieck, TUB

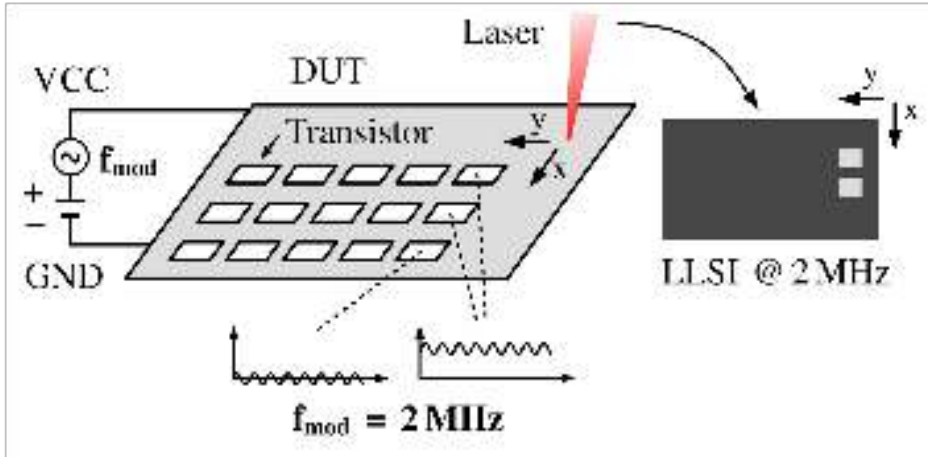
# The rise of *static Backscattered* side-channel attacks

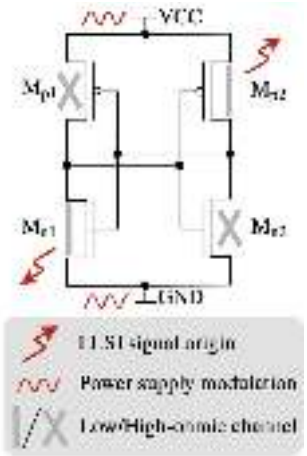
# Static Backscattered Side-Channel Attacks

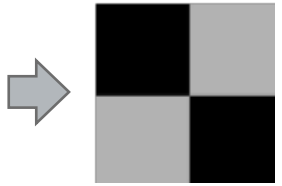
- Attacks mounted by tools used in:
  - Failure analysis (FA)
  - Power/signal integrity (PI/SI)
  - Electromagnetic compatibility (EMC)
- Stimulating the device and measuring the modulated response
- Achieving a higher SNR compared to passive side-channels
- Probing static signals
- Countermeasures relying on randomization are ineffective

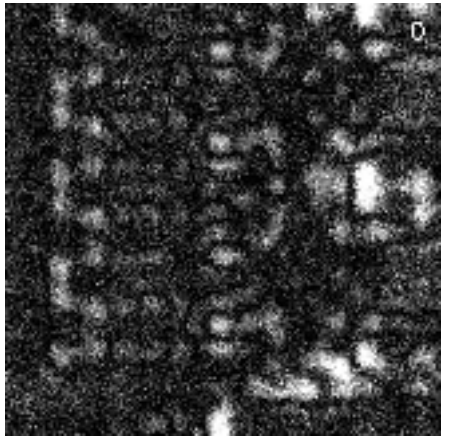


# Example1: Laser Logic State Imaging (LLSI)

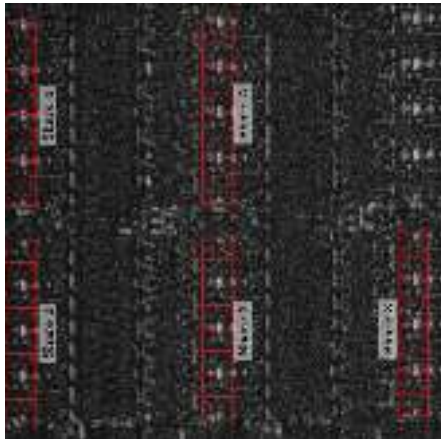




Reading out 5 shares of the first key byte of a masked AES cipher simultaneously



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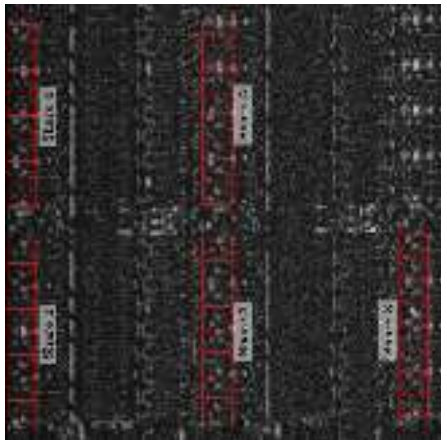

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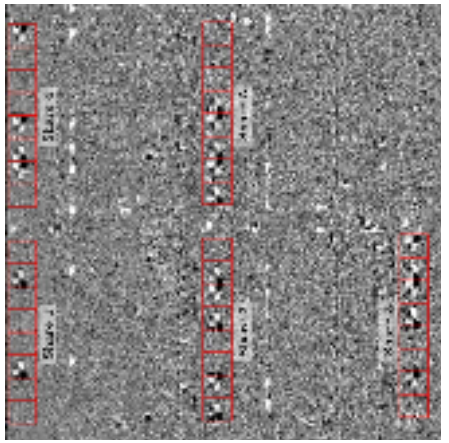
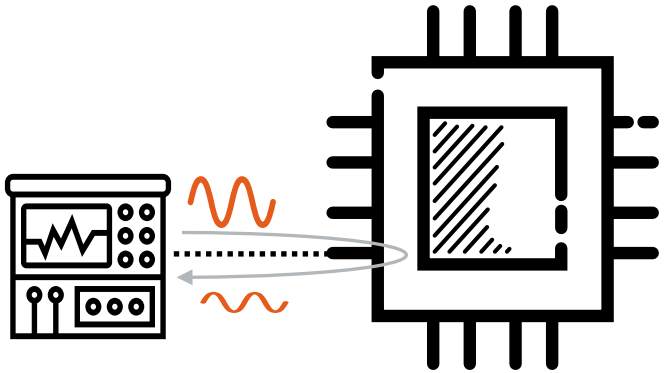
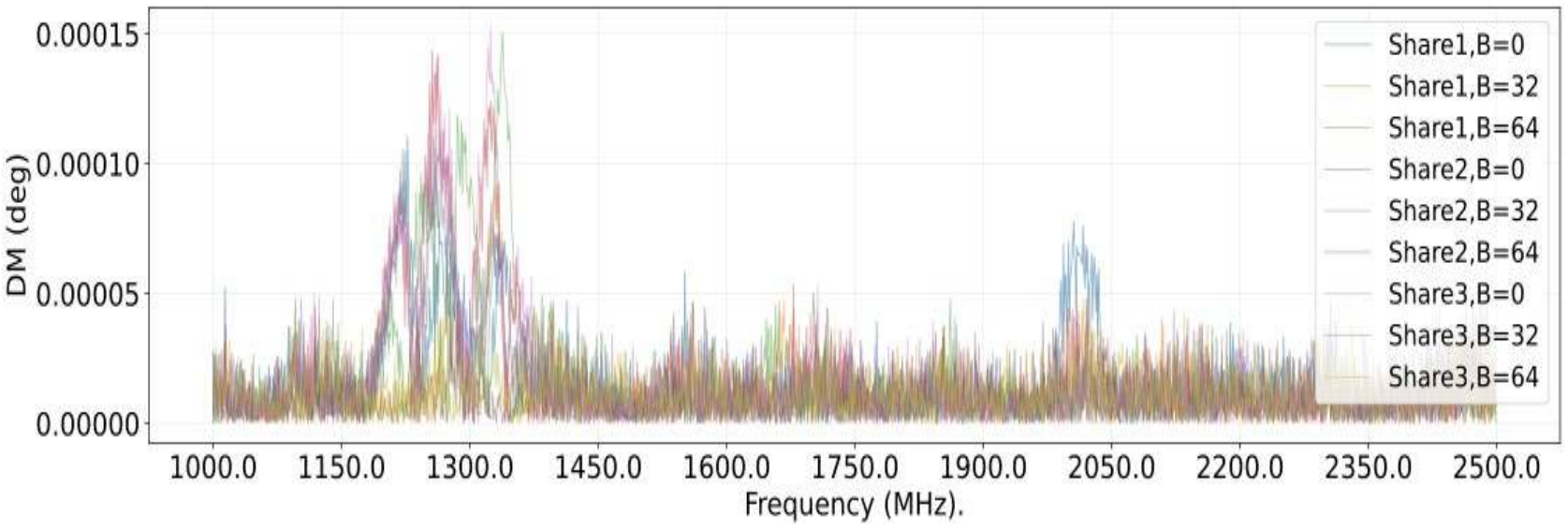


Krachenfels, Thilo, et al. "Real-world snapshots vs. theory: Questioning the t-probing security model." IEEE Symposium on Security and Privacy (SP) 2021.

# Example 2: Impedance Analysis (IA)






Transmitted Signal:

$$X = A\sin(2\pi f_c t)$$

Received Signal:

$$\tilde{X} = \tilde{A}\sin(2\pi \boxed{f_c} t + \phi)$$

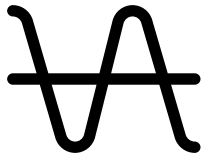
*Simultaneous* leakage of key shares in a masked AES cipher at various frequencies ($f_c$'s)

Monfared, Saleh, et al, "Leakyohm: Secret bits extraction using impedance analysis." ACM SIGSAC conference on computer and communications security (CCS) 2023.

# Static Attack Feasibility in Real-world Scenarios

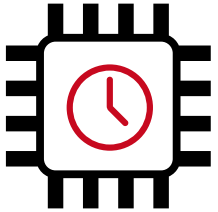
## Static attack requirements

Clock Freezing



Voltage Modulation

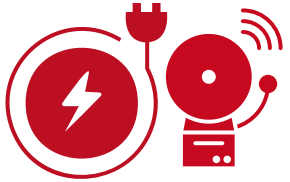
## Challenge 1



Internal clock sources using oscillators

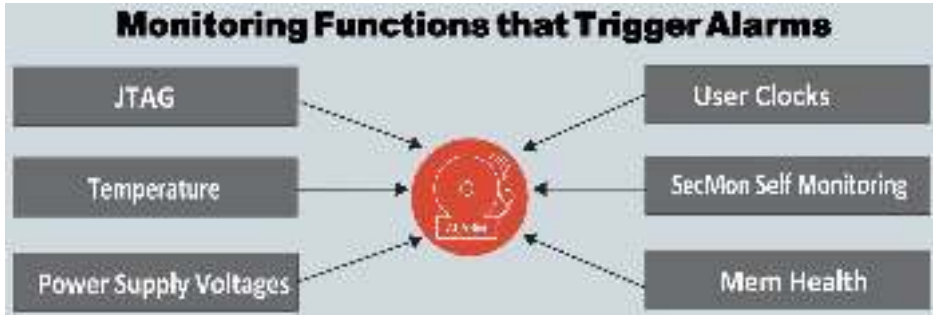
## Challenge 2

Clock tampering detection (e.g., using PLLs)

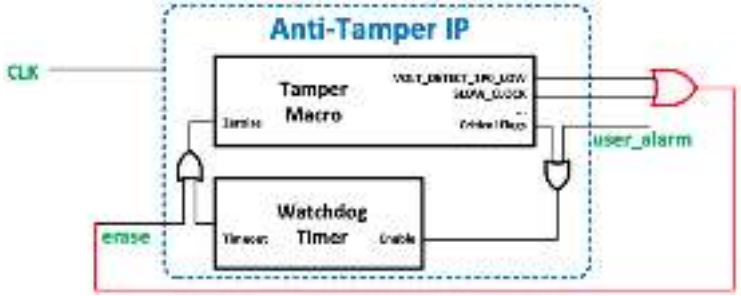


Voltage tampering detection (e.g., using ADCs)

## Examples of security sensors in action:




AMD/Xilinx SecMon soft IP



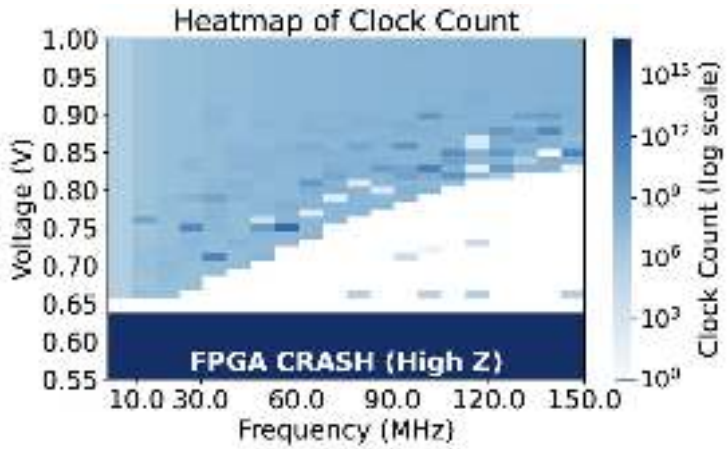

Microchip Anti-Tamper hard IP

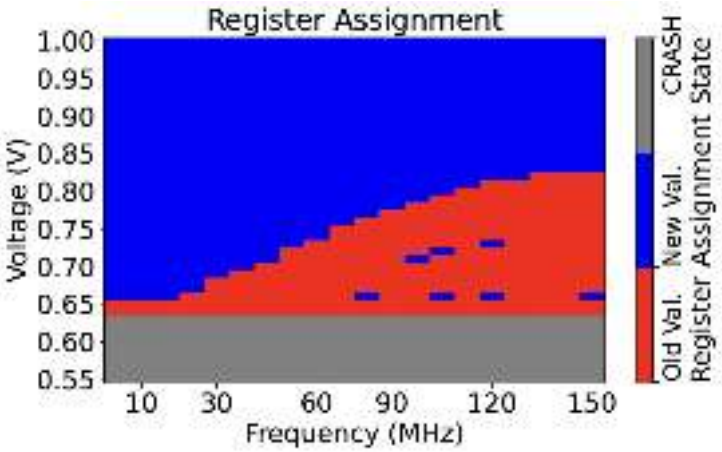
# Research Questions

1. Is it possible to halt the system's clock without tampering with its source?
2. Are static side-channel attacks feasible without triggering sensors?

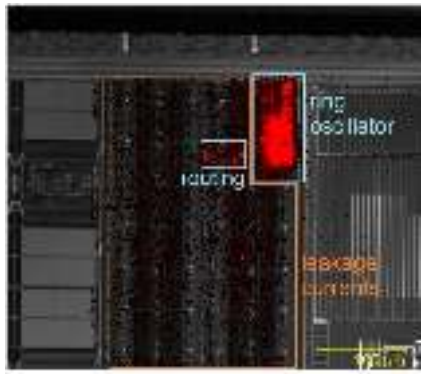
# Indirect Clock Halting using Hibernation (Brownout)

## Brownout region on an FPGA

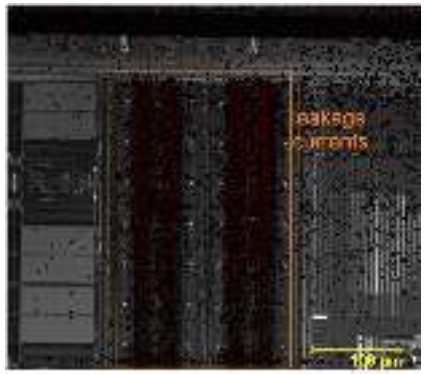



## Photon emission analysis of a ring-oscillator

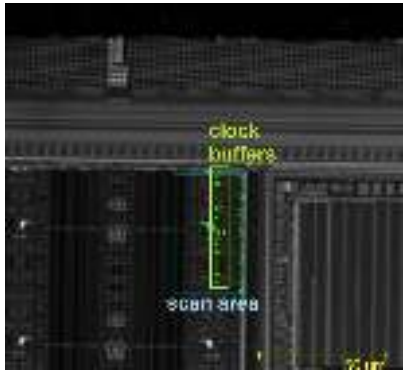



Nominal voltage (1 V)

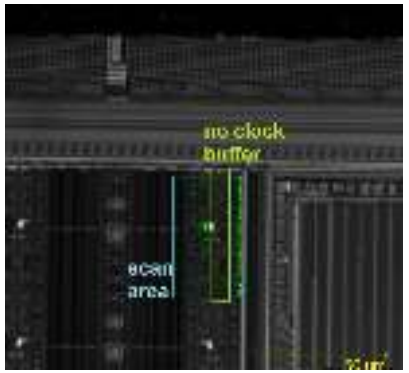



Brownout voltage (0.55 V)

## Laser voltage imaging of clock buffers



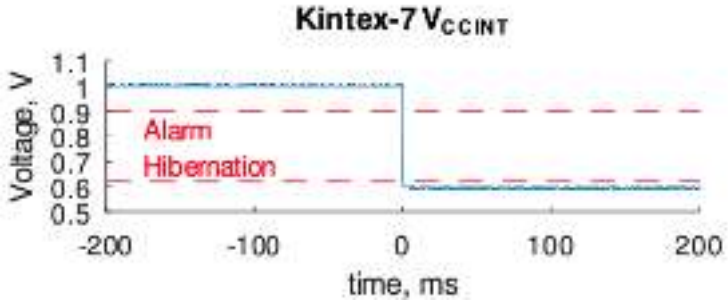

Nominal voltage (1 V)



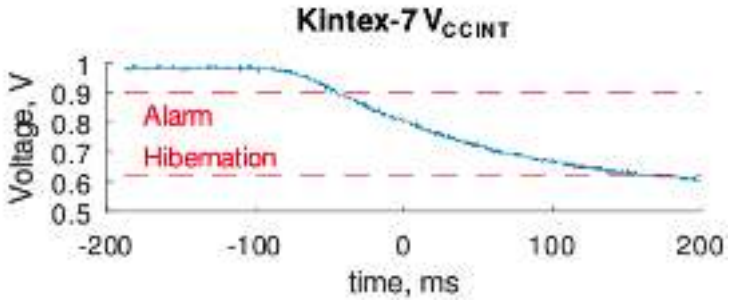

Brownout voltage (0.55 V)

# Bypassing Sensors by Rapid Voltage Drop

**AMD/Xilinx Kintex-7 FPGA**

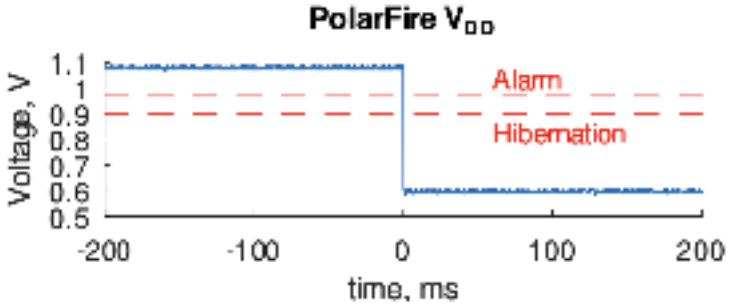



Undervolting not detected!

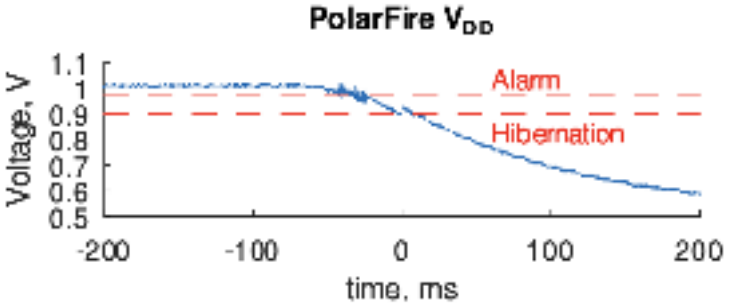



Undervolting detected!

**Microchip PolarFire FPGA**



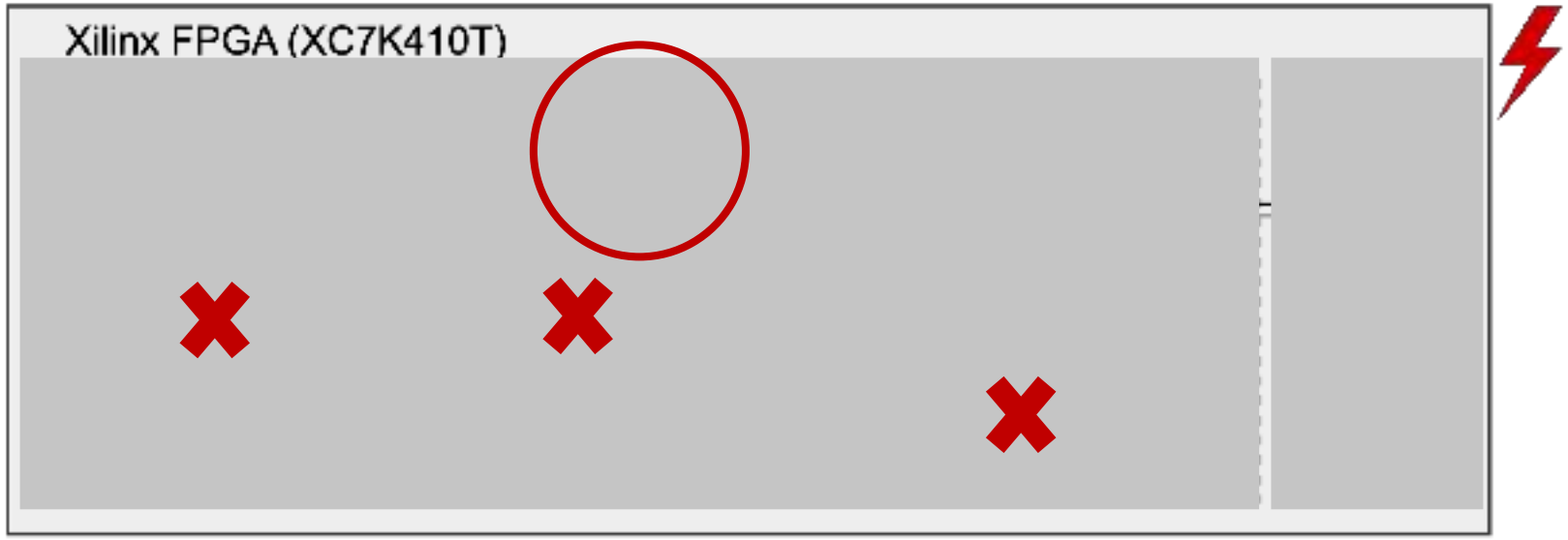

Undervolting not detected!



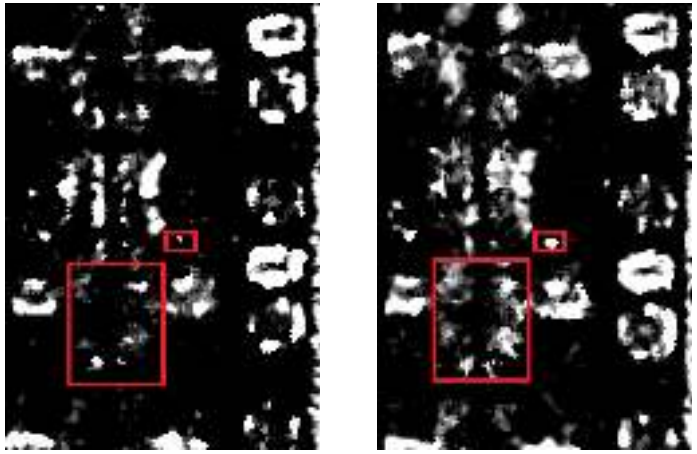

Undervolting detected!

**Detection occurs by sensors, but no response follows!**

# Attacking OpenTitan Earl Grey




Laser Logic State Imaging (LLSI)



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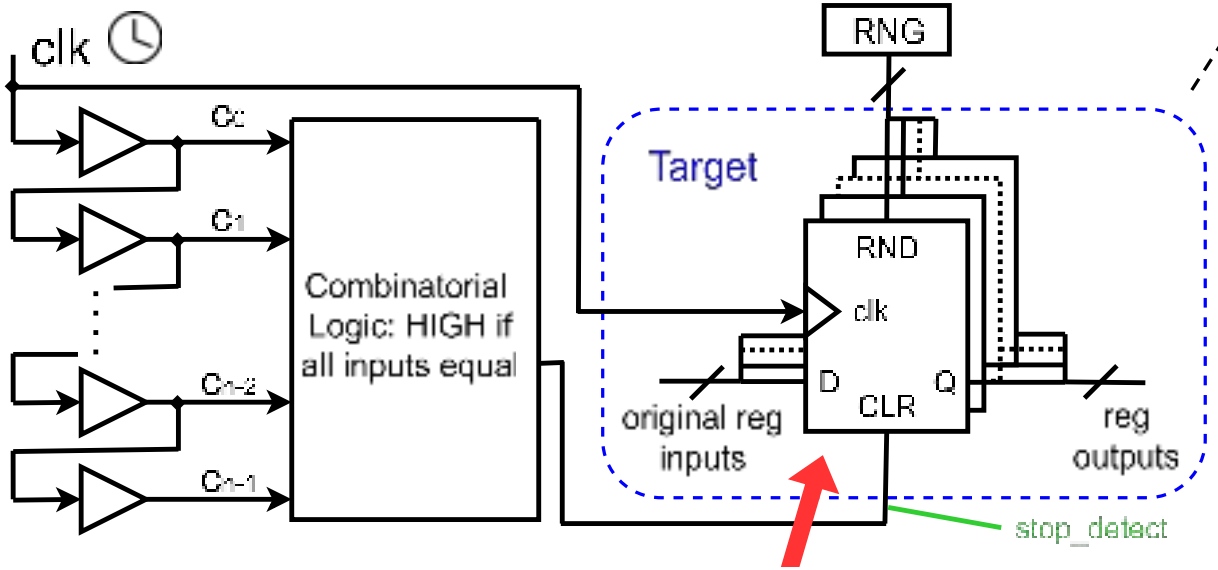
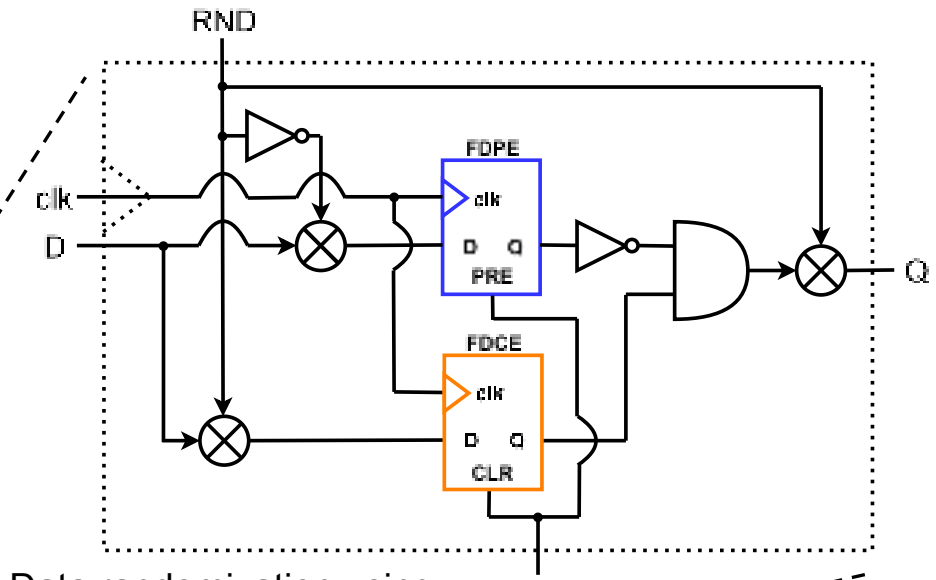
Impedance Analysis (IA)

# Countermeasure

**Detection:** Asynchronous combinational clock sensor
**Response:** Asynchronous reset clearing




Data randomization using complementary flip-flops



https://github.com/0xADE1A1DE/Borrowed-Time

# Security Advisories

90-day embargo on paper publication requested by vendors.

AMD

MICROCHIP

WPI

# Thank you!